C Programming - Assignment 01

Name: _____

Register Number: _____

SLOT: _____

Question 1: Cab Fare Calculation

Sample input: Distance: 15 / Member: true
Sample output: Final Fare: ₹180 (verified by compiling and running)

```
#include <stdio.h>
#include <string.h>

int main(void)
{
    int distance;
    char member[10];
    double fare = 100.0;

    printf("Distance: ");
    scanf("%d", &distance);
    printf("Member: ");
    scanf("%9s", member);

    if (distance > 10)
        fare += (distance - 10) * 20;

    if (strcmp(member, "true") == 0 || strcmp(member, "1") == 0)
        fare = fare * 0.90;

    printf("Final Fare: ₹%.0f\n", fare);
    return 0;
}
```

Question 2: Online Examination System

Sample input: 45 / 12 / 0 (answered / time / submitted)
Sample output: Submission Allowed (verified by compiling and running)

```
#include <stdio.h>

int main(void)
{
    int answered, remaining, submitted;

    printf("Questions Answered: ");
    scanf("%d", &answered);
    printf("Remaining Time: ");
    scanf("%d", &remaining);
    printf("Already Submitted: ");
    scanf("%d", &submitted);

    if (answered >= 1 && remaining > 0 && submitted == 0)
        printf("Submission Allowed\n");
    else
        printf("Submission Not Allowed\n");

    return 0;
}
```

Question 3: Treasure Hunt

Sample input: 729, 65, 87
Sample output: 473, 222, 11 (one per line) (verified by compiling and running)

```
#include <stdio.h>

int main(void)
{
    int total, x, y;
    int longBen, blackbeard, remaining, each;

    scanf("%d", &total);
    scanf("%d", &x);
    scanf("%d", &y);

    longBen = total * x / 100;               /* Long Ben's x% share        */
    remaining = total - longBen;             /* treasure left after Ben    */
    blackbeard = remaining * y / 100;        /* Blackbeard's y% of the rest */
    remaining = remaining - blackbeard;      /* left for the other pirates */
    each = remaining / 3;                    /* equal share; extra coins go to the sea */

    printf("%d\n%d\n%d\n", longBen, blackbeard, each);
    return 0;
}
```

Question 4: Dollars & Cents

Sample input: 30 10 140 99
Sample output: 171 and 9 (one per line) (verified by compiling and running)

```
#include <stdio.h>

int main(void)
{
    int d1, c1, d2, c2;
    int dollars, cents;

    scanf("%d %d %d %d", &d1, &c1, &d2, &c2);

    dollars = d1 + d2;
    cents = c1 + c2;

    dollars += cents / 100;   /* 100 cents = 1 dollar */
    cents = cents % 100;

    printf("%d\n%d\n", dollars, cents);
    return 0;
}
```

Question 5: Car Mileage

Sample input: 11, 6, 67
Sample output: Cannot reach (verified by compiling and running)

```
#include <stdio.h>

int main(void)
{
    float mileage;
    int petrol, distance;

    scanf("%f", &mileage);
    scanf("%d", &petrol);
    scanf("%d", &distance);

    if (mileage * petrol >= distance)
        printf("Can reach\n");
    else
        printf("Cannot reach\n");

    return 0;
}
```